

Telomere-to-telomere assembly using HERRO-corrected Nanopore Simplex reads

<https://doi.org/10.1038/s41586-026-10563-y>

Dominik Stanojević^{1,2}, Dehui Lin¹, Sergey Nurk³, Paola Florez de Sessions³ & Mile Šikić^{1,2✉}

Received: 14 October 2024

Accepted: 17 April 2026

Published online: 27 April 2026

Open access

 Check for updates

Telomere-to-telomere phased assemblies are emerging as a benchmark for reference-quality genomes^{1,2}, although they remain technically and financially demanding, particularly at scale. Generating such assemblies for diploid and polyploid genomes typically involves combining high-accuracy long reads, such as PacBio HiFi³ or the now-deprecated Oxford Nanopore Technologies (ONT) Duplex⁴ reads, with ultra-long ONT Simplex reads. Using multiple platforms or methods increases the cost and the required amount of genomic DNA. Here we show that comparable results are possible using error correction of ultra-long Simplex reads and then assembling them using state-of-the-art de novo assembly methods. To achieve this, we developed the deep learning-based HERRO (haplotype-aware error correction) framework, which corrects Simplex reads while carefully preserving differences in related genomic sequences. Taking into account informative positions that differentiate the haplotypes or genomic repeat copies, HERRO achieves an increase of read accuracy of up to 100-fold for diploid human genomes. By combining HERRO with the Verkko² assembler, we reconstruct up to 32 chromosomes telomere-to-telomere, including chromosomes X and Y, and consistently achieve NGA50 (normalized genome assembly 50) values of 100 Mb or higher across several human genomes. HERRO supports both R9.4.1 and R10.4.1 Simplex reads and generalizes well to other species. These results show that error-corrected ONT reads can lower sequencing costs and improve the quality of genomic analyses.

Both accuracy and contiguity of genome assemblies have seen substantial improvements in recent years owing to advancements in long-read sequencing technologies, new de novo assembly methods and dedicated work of the scientific community on manual curation. Resolving most chromosomes, haplotype-resolved telomere-to-telomere (T2T)¹ diploid assemblies have become a standard.

To achieve haplotype-resolved T2T assemblies, it is currently necessary to use multiple sequencing platforms and methods. These typically include high-accuracy long reads such as HiFi reads produced by Pacific Biosciences sequencers or the now-deprecated Duplex reads^{4,5} produced by ONT. In addition, Simplex ultra-long (UL) reads are often required. Moreover, chromosome-scale phasing usually requires either parental data⁶, single-cell DNA template strand sequencing (Strand-seq⁷) or cross-linking technology (such as Hi-C, Omni-C or Pore-C).

Using multiple platforms and methods can increase both wet lab and dry lab workloads as well as overall project cost. This approach requires a larger amount of high molecular weight genomic DNA, with separate library preparation and sequencing needed for each method. Additionally, more complex algorithms are required to integrate reads from different platforms and methods. Contemporary de novo assembly methods use high-accuracy long reads to construct initial assembly graphs, with lower-accuracy Simplex reads longer than 100 kb (UL reads) serving as auxiliary data to simplify these graphs. Although HiFi

and Duplex reads offer high accuracy, on their own they are usually insufficient for assembling complex regions such as centromeres. They require the addition of UL reads, which, even with their initial lower accuracy, are essential for successfully assembling these challenging regions and achieving haplotype-resolved T2T assemblies. Although Simplex UL read protocols require a substantial amount of input DNA, using them as the sole long-read sequencing method can reduce the total amount of gDNA required for a project, eliminate the need for additional DNA library preparations for other platforms, streamline the sequencing workflow, and unlock the full potential of UL reads.

Several pipelines use error correction of noisy reads, such as Simplex or PacBio continuous long reads, at some stage to produce either collapsed, primary/alt or fully phased, haplotype-resolved assemblies. These include Canu⁸ plus *purge_dups*⁹, FALCON-Unzip¹⁰, Flye¹¹ plus Hap-Dup^{12,13}, Shasta¹⁴ and PECAT¹⁵. However, according to a recent study¹⁵, their performance is inferior compared with strategies that utilize both high-accuracy long reads and Simplex UL reads. For example, using recent Simplex UL data for the HG002 genome, PECAT, the top performer among them, achieved a maximum NGA50 of 91.4 Mb (paternal) and 80.2 Mb (maternal). By contrast, Hifiasm^{3,16,17} and Verkko², utilizing multiple sequencing technologies—HiFi reads, Simplex UL reads and Illumina parental reads—produced assemblies with NGA50 values exceeding 130 Mb (ref. 2 and results herein). They also outperformed PECAT in phasing accuracy, as indicated by a lower Hamming error rate

¹Laboratory of AI in Genomics, Genome Institute of Singapore, A*STAR, Singapore, Singapore. ²Laboratory for Bioinformatics and Computational Biology, Faculty of Electrical Engineering and Computing, University of Zagreb, Zagreb, Croatia. ³Oxford Nanopore Technologies, Oxford, UK. ✉e-mail: mile_sikic@a-star.edu.sg

(2.99% paternal, 1.49% maternal¹⁵ per haplotype versus 0.5% total²). NextDenovo¹⁸, another recently published de novo assembler for noisy long reads, does not phase genomes. Hypo-assembler¹⁹ uses short Illumina reads to correct ONT reads. Cheng et al. extended Hifiasm to support Simplex reads, designating this Hifiasm (ONT)²⁰.

Here, we present HERRO, a framework based on a deep learning model that is capable of correcting Simplex reads. HERRO is optimized for both R9.4.1 and R10.4.1 pores and chemistry and supports both standard and Simplex UL reads.

HERRO achieves up to 100-fold improvement in read accuracy while preserving true biological differences between related genomic sequences, such as variation between haplotypes, segmental duplications or satellite array copies. To accomplish this, we developed a novel deep learning model utilizing convolutional networks and self-attention architecture to process all-versus-all read overlap pileups. Model training has heavily benefitted from the availability of high-quality HG002 genome assembly (<https://github.com/marbl/HG002>).

Combining HERRO with existing de novo assemblers, such as Hifiasm, Verkko and La Jolla Assembler (LJA)²¹, we achieve highly contiguous haplotype-resolved assemblies, with many chromosomes represented as T2T contigs or scaffolds. Despite being trained only on a subset of the human HG002 genome, HERRO performs well across both human and non-human model organisms with diverse genome architectures.

HERRO is a method (Extended Data Fig. 1) and software package aimed at increasing the quality of ONT reads. It utilizes a read correction model that focuses on positions in reads representing differences between haplotypes or specific segments within segmental duplications. Since these positions hold valuable information for phasing, we refer to them as informative positions.

HERRO incorporates a newly developed deep learning model (Fig. 1) that combines convolutional blocks with a Transformer encoder. The model processes windows of stacked bases and quality scores extracted from all-versus-all pairwise alignments (obtained with minimap2²²). Initially, the convolutional blocks process these windows to extract local information from each base and to aggregate information across different reads at the same position, generating representations for each read position. Subsequently, the model selectively processes informative position candidates—defined as those having at least two different bases (including gap symbol) that each appear a minimum of three times—through the Transformer encoder. This encoder is designed to learn interactions between these positions. By learning to predict bases at these challenging positions, HERRO accurately corrects reads while preserving the differences between haplotypes. For the remaining read positions, we use a simple majority voting method to predict the correct bases.

Unlike existing tools such as Racon, DeChat and PECAT, which rely on heuristic-based algorithms, HERRO uses a deep learning model for haplotype-aware error correction of ONT reads. Racon and PECAT use partial-order alignment (POA), which is computationally costly on long reads. PECAT applies hand-crafted rules to select haplotype-consistent reads, but these do not capture read-level interactions or sequence complexity in regions like segmental duplications. Medaka uses a GRU (gated recurrent unit)-based model to polish haploid assemblies from base frequency tables, which lose read-specific context and are limited in modelling long-range dependencies. This makes HERRO a more accurate and robust choice for error correction, particularly in complex or heterozygous regions where traditional methods struggle. A detailed comparison is provided in Supplementary Information 15.

HERRO improves Simplex read accuracy

We evaluated the alignment-based accuracy of the HERRO-corrected reads on several R10 and one R9 ONT datasets with existing high-quality assemblies. In terms of human samples, we considered HG002 data from Human Pangenome Reference Consortium (HPRC) and Genome In a Bottle (GIAB), CHM13 (pseudo-haploid) data from the T2T consortium

and I002C Indian male data with a recently released high-quality assembly (<https://github.com/lbcb-sci/I002C> or <https://github.com/LHG-GG/I002C>). We also used two inbred datasets: *Danio rerio* (zebrafish, TU strain) from the Vertebrate Genomes Project²³ (VGP) and *Arabidopsis thaliana* (thale cress, Col-0 strain)²⁴ (see Data availability). The CHM13 data were sequenced using e8 chemistry and R9.4.1 flow cells. The other datasets were sequenced using e8.2 chemistry and R10.4.1 flow cells.

We aligned both uncorrected and corrected reads to the references using minimap2 and used bamConcordance tool²⁵ to obtain a read quality measure (Q_c) and compute the rates of different types of errors (Methods, 'Analysis methods'). Only primary alignments were considered in the evaluation. For reads successfully processed by HERRO, we observe that HERRO improved Q_c for the majority of reads (Fig. 2a). There is an upward shift in the overall distribution of Q_c values for all considered datasets with an almost 100-fold increase in the median accuracy following correction (Fig. 2b). Extended Data Fig. 2 shows that a similar increase is also observed for higher quality (Q28) HG002 UL dataset.

Table 1 shows the rates of different types of errors in the reads before and after correction. HERRO reduced the rates across all error types, achieving, on average, a 50-fold reduction in the total number of errors across all datasets. Even for the *D. rerio* dataset, which showed the smallest reduction, the error rate has decreased nearly tenfold. The most substantial improvement was observed in mismatch errors, which in human datasets dropped from more than 100 errors per 10 kb to less than 1 error per 10 kb. Similar reductions were observed for other types of errors except for non-homopolymer deletions in the *D. rerio* dataset. The higher error rate for non-homopolymer deletions in this dataset may be due to the prevalence of highly repetitive, low-complexity elements in the genome²⁶.

Notably, centromeric satellites and segmental duplications showed higher error rates after correction (11.8 versus 2.5 per 10 kb in the rest of the genome), driven primarily by non-homopolymer insertion-deletion mutants (indels); mismatch errors were also increased but to a lesser extent (Supplementary Table 13). To assess the risk of overcorrection, we evaluated how often correct bases in the original HG002 reads were erroneously changed during correction. HERRO showed minimal overcorrection genome-wide, with only 1.5 bases per Mb affected when excluding dimers and homopolymers. Overcorrection remained low even in complex regions such as centromeric satellites and segmental duplications (Supplementary Table 14).

Comparison with existing tools

We benchmarked HERRO against six other ONT error-correction methods—DeChat, Canu, Racon, NextDenovo, PECAT and Hifiasm (ONT)—using HG002 Simplex data (Supplementary Tables 13 and 14 and Supplementary Fig. 1). Hifiasm (ONT), DeChat, Canu, Racon and NextDenovo lagged far behind, leaving PECAT as the closest-performing alternative. Notable differences between HERRO and PECAT appear in the most challenging sequence contexts—centromeric satellites and segmental duplications—where HERRO reduces the total error rate to 11.8 per 10 kb, compared with 13.2 per 10 kb with PECAT (Supplementary Table 13). Overcorrection is the clearest separator: compared with PECAT, HERRO introduces about 2× fewer errors overall, 5× fewer after excluding dimers and homopolymers, and more than 7× fewer in complex regions (Supplementary Table 14). Together, these results show that HERRO corrects more accurately while minimizing overcorrection, making it the most reliable choice for Simplex reads—especially in complex regions.

Additional analysis

Corrected reads performed similarly or better in structural variant recall compared with uncorrected reads (Supplementary Table 15), suggesting that heterozygous variants were preserved, although more false positives were called, possibly owing to inconsistent alignments in repetitive regions (Supplementary Information 12 and Supplementary

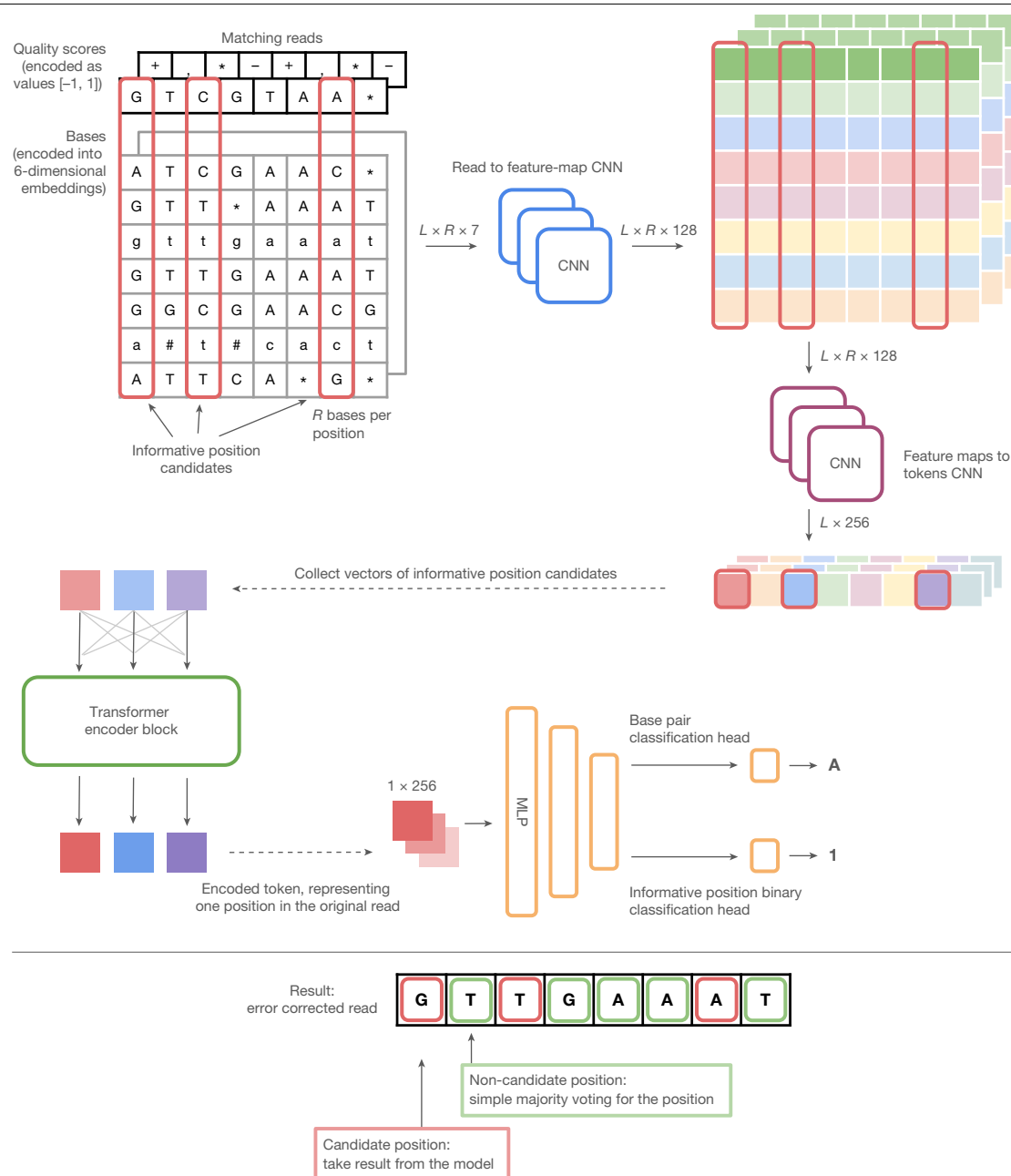


Fig. 1 | The HERRO deep learning model. The input to the model consists of embedded bases concatenated with base qualities, obtained by stacking pairwise alignments between a target read and the reads aligned to it. The first convolutional block extracts local information for each read independently. The second convolutional block aggregates information from different reads at a specific position in the target read. Next, only informative position

candidates are indexed and fed into the Transformer encoder. The output from the Transformer encoder is used for base prediction. Bases at other target read positions are predicted using majority voting. The informative position prediction task is used for model training. CNN, convolutional neural network; MLP, multi-layer perceptron.

Fig. 3). Coverage loss after correction was minimal: excluding regions with pre-existing low coverage, low-coverage intervals spanned just 0.24 Mb, compared with more than 5.3 Mb for PECAT (Supplementary Table 19). Correction also improved homopolymer accuracy by more than 30% (Supplementary Information 14 and Supplementary Figs. 4 and 5).

HERRO enables T2T-grade assemblies

Next, we evaluated the performance of HERRO-corrected ONT reads in a de novo assembly setting using several assemblers: Hifiasm, Verkko and LJA (only for datasets with very low heterozygosity). In addition to

the datasets used to measure read accuracy in the previous section, we included three additional datasets: HG02818 and HG01099 from HPRC, HG005 from GIAB and a *Drosophila melanogaster* dataset (see Data availability). All datasets were sequenced using e8.2 chemistry and R10.4.1 flow cells except for HG01099 and CHM13 which used e8 chemistry and R9.4.1 flow cells. Assembly results for diploid human genomes assembled with HERRO-corrected reads using Verkko as an assembler in combination with parental data are presented in Table 2 (with full results including other assemblers in Supplementary Tables 1–4 and 8), while results for low-heterozygosity genomes (CHM13, zebrafish, *Drosophila* and *Arabidopsis*) are presented in Extended Data Table 1 (with full results in Supplementary Tables 5–8).

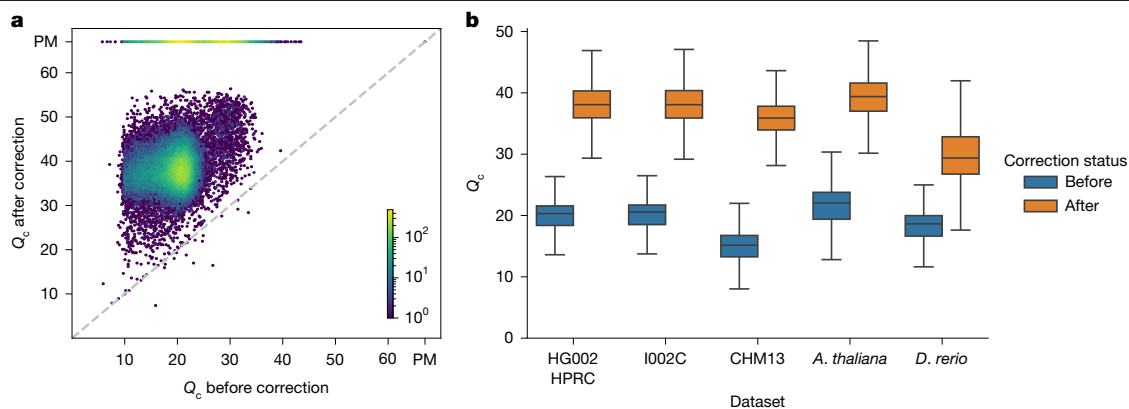


Fig. 2 | Q_c before and after correction. **a**, A hexbin plot representing the empirical joint distribution of Q_c values before and after correction for reads aligned to paternal and maternal chromosome 9 of the HG002 assembly v1.0.1^{1,2,22–24} ($n = 116,166$ reads). The colour indicates the density of data points in each hexagonal bin on a logarithmic scale. Bins above the dashed diagonal line indicate an improvement in accuracy after correction, whereas bins below indicate a reduction. The ‘PM’ at the end of each axis represents a ‘perfect match’. **b**, Box plot showing the distribution of Q_c before and after correction across multiple datasets (number of reads per dataset: HG002 HPRC, $n = 1,770,134$; I002C, $n = 2,716,544$; CHM13, $n = 3,302,229$; *A. thaliana*,

$n = 219,778$; *D. rerio*, $n = 2,037,020$). I002C is a male Indian sample with a recently released reference. For all datasets, alignments that perfectly match the reference were excluded. Chromosomes of HG002 used to train the HERRO model were excluded (1, 2, 17, 18 and 20). CHM13 data was sequenced using R9.4.1 pore. All datasets shown here, except for *A. thaliana*, were sequenced with a Simplex UL kit. Data in the box plots are displayed as follows: the centre line indicates the median, the bounds of the box represent the first and third quartiles, and the whiskers extend to the minimum and maximum values within 1.5 times the interquartile range. Outliers are not shown owing to the large number of data points.

For diploid human datasets, we also used available parental Illumina reads to enable the ‘trio analysis’ modes in Hifiasm and Verkko. HERRO-corrected reads were provided as high-accuracy long inputs, while original reads were provided as UL inputs. Corrected reads were downsampled to around 35× for assembly if their coverages were 38× or above to control the effect of input coverage on assembly results and reduce computational requirements. Corrected reads were cut into 30 kb subreads (minimum 10 kb) for Hifiasm to mitigate contained read exclusion issues^{27,28}; LJA used only the corrected reads as input (Methods). Key assembly metrics across selected assemblies are summarized in Table 2. For more details regarding the different runs, please refer to Supplementary Tables 1–8.

Diploid human genomes

We assembled multiple diploid human genomes, including HG002, I002C, HG02818, HG005 and HG01099, using HERRO-corrected reads (Table 2). For R10.4.1 UL ONT datasets, more than half of the 46 chromosomes were resolved as T2T contigs or scaffolds. Assemblies of HG002 and I002C using HERRO v0.1.1 reached more than 30 T2T chromosomes. Across datasets, when HERRO-corrected reads were used as input, Verkko assemblies achieved more T2T chromosomes and greater contiguity, whereas Hifiasm assemblies reached higher quality value (QV) in some cases (Supplementary Tables 1–4). For HG002, HERRO-corrected reads yielded assemblies with higher NGA50, a greater number of T2T chromosomes, and lower switch and Hamming error rates than those generated from HiFi or Duplex data (Table 2). The slightly larger assembly size observed in the Verkko assembly is due to short contigs mapping to repetitive regions such as ribosomal DNA (rDNA) and various satellites (Supplementary Information 13).

To benchmark HERRO assemblies against those generated from uncorrected reads or other correction approaches, we also evaluated a range of alternative tools on the HG002 dataset, including Hifiasm (ONT), PECAT, DeChat²⁹, Racon, NextDenovo, Canu and Hypo-assembler. Uncorrected reads (that is, reads provided without external correction, relying solely on the integrated internal correction of the assembler; Supplementary Information 10 and Supplementary Tables 2A, 4A, 6A, 7A and 8A) or reads corrected with DeChat, Racon, NextDenovo and Canu, as well as assemblies produced with

Hypo-assembler, either failed to complete or resulted in poor-quality assemblies (Supplementary Table 2A). Among the remaining tools, PECAT produced moderate-quality assemblies that improved when paired with Hifiasm or Verkko rather than its default pipeline; we used this set-up for comparisons and also applied it to I002C (Supplementary Table 3A), but PECAT-based assemblies remained worse than HERRO-based ones. Hifiasm (ONT) (v0.25) was the only competing approach of sufficient quality to warrant evaluation across all datasets. It produced assemblies with equal or in some cases superior contiguity, as reflected by NG50 and NGA50 values across human samples. However, HERRO-based assemblies consistently achieved superior base-level accuracy: Merqury QV improved by 0.3–4.7 points across haplotypes in HG002 and HG02818, alongside reductions in both switch and Hamming error rates. Notably, Hifiasm (ONT) exhibited pronounced haplotype-asymmetric base-level errors on some samples: in HG002, the maternal haplotype showed an approximately 6× higher Merqury Hamming error rate than the paternal haplotype (0.145% versus 0.025%), an asymmetry that is reduced in HERRO-based assemblies (0.040% versus 0.017%). Notably, the higher Hamming error rate relative to switch-error rate in some of the Hifiasm (ONT) assemblies suggests that the contiguity gains come partly at the cost of incorporating long misassigned blocks from the opposing haplotype. Full results regarding Hifiasm (ONT) assemblies can be found in Supplementary Tables 1A–8A. A full comparison with other correction methods is available in Supplementary Information 9.

In multiple runs, the X and Y chromosomes were assembled as T2T contigs or scaffolds, with some runs achieving T2T contigs for both. Whereas Hifiasm successfully assembled the X chromosome as T2T in some runs, only Verkko managed to assemble the challenging³⁰ chromosome Y as T2T. These contigs were evaluated against the X and Y chromosomes from HG002 assembly v1.0.1 using QUAST³¹ (Methods, ‘Analysis methods’), achieving genome fractions of 100% and 99.98% for X and Y, respectively, with a total of 7 misassemblies (Extended Data Table 2).

We evaluated the effect of coverage on correction and assembly quality using the I002C dataset. Assembly contiguity and T2T chromosome count increased consistently with the total uncorrected dataset coverage available for correction: at 71× correction coverage (after

Table 1 | Counts of errors before and after correction, by type

Dataset	Coverage (raw, preprocessed, downsampled, corrected)	Correction	Mismatches per 10 kb	Non-HPIs per 10 kb	Non-HPDs per 10 kb	HPIs per 10 kb	HPDs per 10 kb	Errors per 10 kb
HG002	76, 66, 66, 65	Before	133.15	45.56	58.54	26.22	71.02	334.48
		After	0.23	0.63	0.74	0.41	1.77	3.78
I002C	79, 63, 63, 63	Before	186.86	58.76	79.74	33.41	99.44	458.21
		After	0.22	1.01	1.72	0.50	2.18	5.64
HG005	76, 67, 67, 66	Before	112.97	37.50	45.80	23.39	60.23	279.89
		After	0.10	0.16	0.29	0.34	1.45	2.33
HG02818	70, 57, 57, 56	Before	182.05	57.44	76.09	32.92	96.82	445.31
		After	0.22	1.05	1.07	0.41	2.07	4.81
HG01099	102, 67, 67, 66	Before	318.96	116.99	140.21	60.95	184.95	822.07
		After	0.52	0.38	0.96	2.14	1.55	5.54
CHM13	82, 67, 67, 65	Before	158.63	74.75	91.23	46.91	128.49	500.01
		After	0.38	0.34	0.85	1.23	1.60	4.39
<i>A. thaliana</i>	223, 160, 80, 59	Before	38.59	27.84	22.82	10.03	29.09	128.37
		After	0.70	1.37	1.92	0.05	2.85	6.89
<i>D. rerio</i>	134, 125, 80, 79	Before	71.84	29.64	50.04	17.27	41.64	210.43
		After	0.41	2.23	15.15	0.10	3.88	21.77

We used bamConcordance to obtain counts of errors in reads before and after correction, based on alignments to a high-quality reference. In addition to the total number of errors, counts are provided for different categories: mismatches, non-homopolymer insertions (non-HPIs), non-homopolymer deletions (non-HPDs), homopolymer insertion (HPIs) and homopolymer deletion (HPDs). The table additionally reports coverage at key processing steps: raw data, after preprocessing (trimming (if applicable) and quality and length filtering), after downsampling (for high-coverage non-human datasets), and after correction. The HERRO version for this analysis was v0.1.

length and quality filtering), both haplotypes reached NG50 of 146 Mb, compared to 90–130 Mb when correction coverage was around 35× (Supplementary Tables 1–3). To isolate the effect of correction coverage from assembly coverage, we held both the high-accuracy assembly input (corrected reads downsampled to 35×) and the UL assembly input (uncorrected reads at 56×) fixed while varying only the coverage used for correction (45×, 63× and 71×). The number of T2T contigs or scaffolds increased from 15 to 21 to 28 as correction coverage increased, and yak QVs followed the same trend (Supplementary Tables 3 and 9). Whole-genome assemblies were also evaluated (Supplementary Table 16), including stratified analyses by repeat status (Supplementary Table 17). Robustness to basecalling model choice was also assessed by re-basecalling HG002 reads with the hac and fast models (Supplementary Information 11, Supplementary Fig. 2 and Supplementary Table 2B).

To test on non-UL human data, we corrected and assembled the HG005 dataset. Although this is not a UL dataset, it still has 10× coverage of reads that are 50 kb or longer. In addition to the ONT reads, parental data was also provided to assemblers. When assembled with Verkko, 23 T2T chromosomes were obtained, including three that are T2T in single contigs. Full details about this run, as well as other runs for the HG005 dataset, can be found in Supplementary Table 4.

Finally, we corrected and assembled experimental higher-accuracy (Q28) UL HG002 reads using a model trained on standard R10 data. Verkko produced 11 T2T contigs and 14 T2T scaffolds with the corrected reads but produced none without correction. Hifiasm was able to produce 1 T2T contig and 18 T2T scaffolds without correction, and 7 T2T contigs and 12 T2T scaffolds with correction. The QV for each haplotype increased by around 10 with correction. These results demonstrate that error-correcting highly accurate data improves assembly contiguity, resulting in more T2T chromosomes in contigs with higher accuracy (Supplementary Table 10).

Difference between genomes

While evaluating error rates and assembly quality, we noticed discrepancies in the results depending on the reference genome used in the evaluation. We thus decided to quantify the differences in single

nucleotide variants (SNVs) and indels (≤50 bp) across the different genomes.

Extended Data Table 3 shows aggregated error rates of corrected reads evaluated by mapping to different high-quality assemblies (HG002 v1.0.1, CHM13 v2.0, CN1 v0.8³² and I002C v0.4). Since the error rate of corrected reads is very low, it can be almost neglected. The results show that differences in mean mismatch and indel rate are 0.15–0.19% and 0.14–0.19%, respectively. In addition, we see notable differences in the NGA50 measure when evaluating genome assemblies against different references, with differences possibly exceeding 20% (Supplementary Table 12). It is important to note that notable variations persist even when another haplotype from the same genome is used as a reference.

These results highlight the importance of using high-quality assemblies and reads originating from the same genome when training deep learning models. Since the error rate of mapping reads to a high-quality assembly from the same individual is low, we hypothesize that mapping highly accurate reads from one individual to a high-quality assembly of another can provide a rapid estimate of the differences between two genomes. Notably, the mean percentage of mismatches remains stable across genome pairs, offering an estimate of the number of SNVs. The observed range of 0.15–0.19% is higher than some previous estimates of 0.1%³³. However, it is consistent with more recent findings³⁴, where authors reported that the percentage of SNVs in segmental duplication regions is around 0.15%. Moreover, a recent study³⁵ suggests that there are similar differences between genomes for both SNVs and indels, with indels not having been quantified in previous studies.

Low heterozygosity non-human genomes

We corrected and assembled ONT R10 data of highly homozygous samples for three non-human organisms: *A. thaliana* (Col-0 strain), *D. melanogaster* (iso-1 strain), and *D. rerio* (TU strain), with only *D. rerio* being a UL dataset. For *A. thaliana*, Hifiasm assembled three out of five chromosomes as T2T contigs (Extended Data Table 1). LJA outperformed Hifiasm and Verkko in assembling *D. rerio*, achieving four T2T contigs compared to zero and one, respectively (Supplementary Table 5). When we used *D. rerio* corrected reads longer than 90 kb for

Table 2 | Assembly accuracy and T2T continuity for diploid human genomes

Assembly	Haplotype	Size (Gb)	NG50 (Mb)	NGA50 (Mb)	QV	Hamming errors, switch errors (%)	Single-copy genes lost, single-copy genes duplicated, multi-copy genes lost (%)	Number of T2T contigs (plus scaffold)
HG002, ONT (HPRC), Verkko, HERRO	Maternal	3.115	154	135	54.63	0.04, 0.02	0.68, 0.26, 5.19	10 (16)
	Paternal	2.995	136	134	56.55	0.02, 0.02	0.79, 0.23, 7.83	12 (16)
HG002, HiFi plus Simplex UL, Hifiasm, uncorrected Cheng et al. ³	Maternal	3.026	112	110	56.50	0.07, 0.04	0.91, 0.21, 5.29	4 (5)
	Paternal	2.946	134	133	55.95	0.04, 0.03	0.92, 0.30, 7.98	3 (10)
HG002, Duplex plus Simplex UL, Verkko, uncorrected Koren et al. ⁴	Maternal	3.037	147	114	53.59	0.05, 0.03	0.78, 0.34, 5.19	6 (11)
	Paternal	2.943	134	134	53.23	0.03, 0.02	0.85, 0.26, 8.03	10 (15)
HG002, ONT (HPRC), Hifiasm (ONT), uncorrected	Maternal	3.056	154	144	54.29	0.14, 0.03	0.93, 0.52, 5.39	10 (15)
	Paternal	2.945	147	134	55.00	0.03, 0.02	0.82, 0.27, 7.98	17 (19)
IO02C, ONT, Verkko, HERRO	Maternal	3.057	155	135	60.50	0.27, 0.25	0.55, 0.21, 4.54	10 (16)
	Paternal	2.989	135	108	58.53	0.18, 0.17	0.82, 0.34, 6.24	6 (15)
IO02C, ONT, Hifiasm (ONT), uncorrected	Maternal	3.048	155	133	58.30	0.35, 0.26	0.72, 0.41, 4.59	11 (16)
	Paternal	2.951	140	105	58.76	0.21, 0.20	1.06, 0.59, 6.79	11 (16)
HG02818, ONT, Verkko, HERRO	Maternal	3.097	154	92	58.59	0.032, 0.030	0.72, 0.33, 4.44	7 (16)
	Paternal	3.044	137	111	58.93	0.034, 0.022	0.76, 0.34, 7.09	10 (16)
HG02818, ONT, Hifiasm (ONT), uncorrected	Maternal	3.056	154	98	53.91	0.204, 0.038	0.72, 0.32, 4.70	16 (19)
	Paternal	3.029	154	107	54.81	0.033, 0.032	0.77, 0.31, 7.19	15 (17)
HG005, ONT (non-UL), Verkko, HERRO	Maternal	3.045	135	88	57.58	0.022, 0.017	1.12, 0.47, 6.79	1 (12)
	Paternal	2.955	135	99	57.92	0.011, 0.010	0.74, 0.33, 7.44	2 (11)
HG005, ONT (non-UL), Hifiasm (ONT), uncorrected	Maternal	3.038	135	96	56.77	0.105, 0.085	0.87, 0.27, 6.79	9 (15)
	Paternal	3.001	135	103	55.97	0.063, 0.047	0.74, 0.37, 7.64	9 (14)
HG01099, ONT (R9.4.1), Verkko, HERRO	Maternal	3.199	97	71	45.00	0.407, 0.404	2.82, 2.00, 8.88	0 (6)
	Paternal	3.090	87	75	43.02	0.233, 0.235	2.62, 2.09, 5.04	0 (3)
HG01099, ONT (R9.4.1), Hifiasm (ONT), uncorrected	Maternal	2.971	40	30	38.17	2.424, 0.378	6.70, 1.38, 13.02	0 (0)
	Paternal	2.873	40	34	37.95	1.568, 0.221	6.30, 1.36, 13.42	0 (0)

For each sample and correction method, we report NG50, NGA50, QV, Hamming- and switch-error rates, gene completeness and duplication, and counts of T2T contigs or scaffolds. T2T continuity for HG002 and IO02C is assessed against high-quality sample-specific references; for HG02818, HG005 and HG01099, T2T chromosome counts are inferred using CHM13 as a reference. The first column specifies the sample, input read type, assembler and error-correction method applied; rows labelled Hifiasm (ONT) use the integrated internal correction in Hifiasm, and rows labelled HERRO use explicit pre-assembly correction with HERRO v0.1.1 prior to assembly with Verkko. All assemblies were generated in trio mode using parental Illumina data for haplotype phasing. Assemblies based on HERRO-corrected reads used HERRO v0.1.1. Assemblies from Cheng et al.³ (HiFi plus UL) and Koren et al.⁴ (Duplex plus UL) provide HG002 benchmarks; Hifiasm (ONT) v0.25 is shown for comparison. See Methods and Supplementary Tables 1–4 and 8 (full results for HERRO-corrected read assemblies) and Supplementary Tables 1A–4A and 8A (full results for assemblies not using HERRO-corrected reads) for tool versions, coverage details, telomere criteria and run-specific notes.

assembly with LJA, the number of T2T contigs increased from 4 to 7 (Supplementary Table 5).

R9.4.1 data

To test the ability of HERRO to correct R9.4.1 Simplex reads, we corrected and assembled a UL dataset from the CHM13¹ genome and HG01099, a diploid male sample. Across the different assemblers used, we achieved an NG50 of approximately 90 Mb and up to 5 chromosomes as T2T contigs for CHM13 (Supplementary Table 8). Hifiasm and LJA produced similar results, with Hifiasm yielding the highest number of T2T chromosomes and LJA producing more complete genes, as assessed using asmgene.

For HG01099, Verkko generated nine T2T scaffolds (no T2T contigs) while Hifiasm produced one T2T contig and one T2T scaffold. The contig NG50 values reached only about 40 Mb compared with 90 Mb for CHM13. This difference is likely to reflect the increased complexity of diploid genome assembly. Full results for all assemblies can be found in Supplementary Tables 8 and 8A.

Runtime assessment

Comprehensive runtime assessments for all datasets are presented in Supplementary Tables 1–8. For the HG002 dataset, with 66× input

coverage, the correction process required approximately 69 h of wall clock time in total. The majority of the time, about 64 h, was spent running minimap2 to obtain all-versus-all alignments, using 64 threads and peaking at 102 GB of RAM. Following this, the correction phase took approximately 5 h with peak RAM consumption of 550 GB using 64 threads. More details can be found in the Methods.

Discussion

Long high-accuracy HiFi and Duplex reads have created new opportunities for reconstructing large and complex genomes. Their low error rates have enabled the assembly of challenging regions, such as segmental duplications, by distinguishing tiny variations between each segment. Moreover, these reads and new assembly methods considerably improved haplotype resolution. However, to reconstruct the most complex regions, such as centromeres and other satellite arrays, and to fully phase the haplotypes, Simplex UL reads are required. Owing to the higher error rate, their usage as a basis for the construction of high-resolution assembly graphs was limited, leading to a hybrid strategy in which an assembly graph is first built from long high-accuracy reads, with further integration of Simplex UL data. This approach combined with trio analysis and manual curation, has

led to several breakthroughs. These include the T2T reconstruction of a pseudo-haploid human genome, the reconstruction of complete human chromosome Y, and the T2T haplotype-resolved assembly of the HG002 diploid human genome.

We began our work with the goal of fully leveraging the potential of Simplex UL reads, thereby removing the need for HiFi or Duplex reads in assembly projects. To achieve this, we decided to try reducing the error rate of ONT reads through self-correction. However, heuristic-based approaches, such as those used in Racon³⁶, PECAT¹⁵, Hifiasm¹⁶ or HiCanu³⁷, have proved to be inadequate as they may overcorrect important, informative positions that are crucial for reconstructing segmental duplications and phasing polyploid genomes. Therefore, we set out to develop a new error-correcting method focused on those key positions.

Using HERRO, a deep learning model trained on Simplex reads labelled with the high-quality HG002 assembly, enables us to correct reads to a level where they can be used in existing assemblers as long accurate reads, without overcorrecting informative positions. The results indicate that we can achieve high contiguity comparable to approaches that combine HiFi or Duplex with Simplex UL reads. However, we believe the current assembly results are not yet optimal owing to several factors, including potential under- or overcorrection by HERRO and the need for assemblers to be further tailored for high-quality UL reads. Current assemblers were largely designed for reads with per-read accuracy in the Q20–Q30 range, such as HiFi reads, and may not fully exploit the higher accuracy of HERRO-corrected reads, which can reach average read quality up to Q40. In our datasets, we were able to reconstruct complete human chromosomes X and Y using Simplex UL reads, suggesting the feasibility of this approach. With continued improvements in error correction and assembler optimization, it may become feasible to routinely assemble even the most complex genomic regions—including rDNAs and polyploid chromosomes—without relying on high-accuracy long reads.

The subsequent development of Hifiasm (ONT)²⁰ provides further evidence that ONT reads alone are sufficient for high-quality, near-T2T genome assembly. HERRO-corrected reads achieve higher per-read accuracy compared to the reads corrected by Hifiasm (ONT), including in homopolymer regions that are particularly challenging for nanopore sequencing (Supplementary Table 13). Comparing our assemblies against Hifiasm (ONT), we find consistent improvements in per-base sequence quality across all samples, as reflected in Merqury QV scores. Notably, Hifiasm (ONT) exhibits a pronounced maternal–paternal asymmetry in Merqury Hamming error rate on some of the human samples, whereas our pipeline reduces this asymmetry across all samples. Contiguity is broadly comparable for most samples, though Hifiasm (ONT) recovers more T2T chromosomes in human assemblies, possibly benefiting from the tight co-design of its error correction and assembly algorithms, including a built-in overlapper optimized for high-quality HiFi or ONT reads. Both pipelines perform less well on HG005, which lacks UL reads, highlighting that read length remains an important factor regardless of assembly strategy. Additionally, unlike Hifiasm (ONT), our pipeline supports R9.4.1 chemistry, broadening its applicability to datasets generated on older instruments. Together, these results suggest that the two approaches occupy complementary positions in the quality–contiguity trade-off space, with HERRO offering a clear advantage in per-base accuracy and haplotype consistency.

Our results indicate that mapping long, highly accurate reads to a high-quality assembly from another individual results in differences of 0.15–0.19% in mismatches and 0.14–0.19% in short indels. Additionally, we found that the structural concordance metrics, such as NGA50, can vary depending on the reference genome used for assessment. These observations highlight the importance of using *de novo* assemblies generated from reads from the same individual rather than relying

on reference sequences built from a different individual. Although we acknowledge that our sample size is small and that a more comprehensive analysis is needed, this demonstrates the potential of using long, highly accurate reads to estimate genomic differences quickly. These insights may provide guidelines for future machine learning methods that rely on read mapping, such as error correction, DNA modification detection, and similar.

We acknowledge room for improvement across the pipeline. Using minimap2 for all-versus-all alignments may be suboptimal in both runtime and accuracy; modern assemblers like Hifiasm and Verkko use built-in overlappers specifically designed for long high-accuracy or UL reads, and developing an overlapper tailored for ONT pre-corrected reads could improve overlap quality while reducing runtime. Promising directions include parallelizing minimap2 on CPU³⁸ or GPU³⁹, modifying existing aligners (such as Edlib⁴⁰ or A*⁴¹) or developing entirely new algorithms. Additional gains could come from better alignment sorting strategies, optimizing window length, and reducing memory requirements.

Manual inspection of residual errors revealed three main types. First, longer homopolymers remain problematic, as ONT reads frequently cannot capture their correct lengths. Second, stacked pairwise alignments can be suboptimal, causing disorganized pileups near homopolymers and short tandem repeats when the target read contains nearby indels. Third, undercorrection occurs when an incorrect base in the target read is well supported and left uncorrected, possibly reflecting conservative model behaviour to avoid overcorrection towards the wrong haplotype. These issues could be mitigated by improving alignment sorting, introducing a consistency loss to encourage agreement at heterozygous sites, and incorporating multiple sequence alignment. HERRO currently handles indels up to 50 nucleotides; support for longer indels warrants exploration.

Beyond these immediate improvements, several broader directions remain open. Training on more data, including non-human organisms, could further reduce error rates. Assembly of metagenomic samples is another promising avenue^{42–44}. The impact of highly accurate ONT reads on haploid assemblers such as Raven⁴⁵ and Flye¹¹ has not yet been explored; while these tools neither reconstruct individual haplotypes nor require UL or parental data, they produce assemblies suitable as references for population studies, and higher input accuracy could improve separation of diverged segmental duplication copies. Finally, incorporating methylation information, for example via Rockfish⁴⁶ or similar tools, may improve the identification of informative positions.

In summary, here we introduce HERRO, an error-correction method that enables high-quality assembly using Simplex reads as the sole long-read sequencing technology. This approach results in lower costs and reduces genomic DNA requirements while achieving state-of-the-art assemblies. Moreover, by providing a model that supports R9.4.1 reads, HERRO enables researchers to use even older data in state-of-the-art assembly pipelines. In future studies, accurate long and Simplex UL reads may facilitate the analysis of more complex genomic architectures, including polyploid and aneuploid organisms, which represent promising directions that are yet to be explored.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-026-10563-y>.

1. Nurk, S. et al. The complete sequence of a human genome. *Science* **376**, 44–53 (2022).
2. Rautiainen, M. et al. Telomere-to-telomere assembly of diploid chromosomes with Verkko. *Nat. Biotechnol.* **41**, 1474–1482 (2023).

3. Cheng, H., Asri, M., Lucas, J., Koren, S. & Li, H. Scalable telomere-to-telomere assembly for diploid and polyploid genomes with double graph. *Nat. Methods* **21**, 967–970 (2024).
4. Koren, S. et al. Gapless assembly of complete human and plant chromosomes using only nanopore sequencing. *Genome Res.* **34**, 1919–1930 (2024).
5. Sarashetti, P., Lipovac, J., Tomas, F., Šikić, M. & Liu, J. Evaluating data requirements for high-quality haplotype-resolved genomes for creating robust pangenome references. *Genome Biol.* **25**, 312 (2024).
6. Koren, S. et al. De novo assembly of haplotype-resolved genomes with trio binning. *Nat. Biotechnol.* **36**, 1174–1182 (2018).
7. Sanders, A. D., Falconer, E., Hills, M., Spierings, D. C. J. & Lansdorp, P. M. Single-cell template strand sequencing by Strand-seq enables the characterization of individual homologs. *Nat. Protoc.* **12**, 1151–1176 (2017).
8. Koren, S., Walenz, B. P., Berlin, K., Miller, J. R. & Phillippy, A. M. Canu: scalable and accurate long-read assembly via adaptive k-mer weighting and repeat separation. *Genome Res.* **27**, 722–736 (2017).
9. Guan, D. et al. Identifying and removing haplotypic duplication in primary genome assemblies. *Bioinformatics* **36**, 2896–2898 (2020).
10. hin, C.-S. et al. Phased diploid genome assembly with single-molecule real-time sequencing. *Nat. Methods* **13**, 1050–1054 (2016).
11. Kolmogorov, M., Yuan, J., Lin, Y. & Pevzner, P. A. Assembly of long, error-prone reads using repeat graphs. *Nat. Biotechnol.* **37**, 540–546 (2019).
12. Shafin, K. et al. Haplotype-aware variant calling with PEPPER-Margin-DeepVariant enables high accuracy in nanopore long-reads. *Nat. Methods* **18**, 1322–1332 (2021).
13. Kolmogorov, M. et al. Scalable Nanopore sequencing of human genomes provides a comprehensive view of haplotype-resolved variation and methylation. *Nat. Methods* **20**, 1483–1492 (2023).
14. Shafin, K. et al. Nanopore sequencing and the Shasta toolkit enable efficient de novo assembly of eleven human genomes. *Nat. Biotechnol.* **38**, 1044–1053 (2020).
15. Nie, F. et al. De novo diploid genome assembly using long noisy reads. *Nat. Commun.* **15**, 2964 (2024).
16. Cheng, H., Concepcion, G. T., Feng, X., Zhang, H. & Li, H. Haplotype-resolved de novo assembly using phased assembly graphs with hifiasm. *Nat. Methods* **18**, 170–175 (2021).
17. Cheng, H. et al. Haplotype-resolved assembly of diploid genomes without parental data. *Nat. Biotechnol.* **40**, 1332–1335 (2022).
18. Hu, J. et al. NextDenovo: an efficient error correction and accurate assembly tool for noisy long reads. *Genome Biol.* **25**, 107 (2024).
19. Darian, J. C., Kundu, R., Rajaby, R. & Sung, W.-K. Constructing telomere-to-telomere diploid genome by polishing haploid nanopore-based assembly. *Nat. Methods* **21**, 574–583 (2024).
20. Cheng, H. et al. Efficient near-telomere-to-telomere assembly of nanopore simplex reads. *Nature* <https://doi.org/10.1038/s41586-026-10105-6> (2026).
21. Bankevich, A., Bzikadze, A. V., Kolmogorov, M., Antipov, D. & Pevzner, P. A. Multiplex de Bruijn graphs enable genome assembly from long, high-fidelity reads. *Nat. Biotechnol.* **40**, 1075–1081 (2022).
22. Li, H. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics* **34**, 3094–3100 (2018).
23. Rhie, A. et al. Towards complete and error-free genome assemblies of all vertebrate species. *Nature* **592**, 737–746 (2021).
24. Wang, B. et al. High-quality *Arabidopsis thaliana* genome assembly with nanopore and HiFi long reads. *Genomics Proteomics Bioinformatics* **20**, 4–13 (2022).
25. Wenger, A. M. et al. Accurate circular consensus long-read sequencing improves variant detection and assembly of a human genome. *Nat. Biotechnol.* **37**, 1155–1162 (2019).
26. Chernyavskaya, Y., Zhang, X., Liu, J. & Blackburn, J. Long-read sequencing of the zebrafish genome reorganizes genomic architecture. *BMC Genomics* **23**, 116 (2022).
27. Kamath, S. S., Bindra, M., Pal, D. & Jain, C. Telomere-to-telomere assembly by preserving contained reads. *Genome Res.* **34**, 1908–1918 (2024).
28. Jain, C. Coverage-preserving sparsification of overlap graphs for long-read assembly. *Bioinformatics* **39**, btad124 (2023).
29. Liu, Y. et al. Repeat and haplotype aware error correction in nanopore sequencing reads with DeChat. *Commun. Biol.* **7**, 1678 (2024).
30. Rhie, A. et al. The complete sequence of a human Y chromosome. *Nature* **621**, 344–354 (2023).
31. Gurevich, A., Saveliev, V., Vyahhi, N. & Tesler, G. QUAST: quality assessment tool for genome assemblies. *Bioinformatics* **29**, 1072–1075 (2013).
32. Yang, C. et al. The complete and fully-phased diploid genome of a male Han Chinese. *Cell Res.* **33**, 745–761 (2023).
33. Pang, A. W. et al. Towards a comprehensive structural variation map of an individual human genome. *Genome Biol.* **11**, R52 (2010).
34. Vollger, M. R. et al. Increased mutation and gene conversion within human segmental duplications. *Nature* **617**, 325–334 (2023).
35. Logsdon, G. A. et al. Complex genetic variation in nearly complete human genomes. *Nature* **644**, 430–441 (2025).
36. Vaser, R., Sović, I., Nagarajan, N. & Šikić, M. Fast and accurate de novo genome assembly from long uncorrected reads. *Genome Res.* **27**, 737–746 (2017).
37. Nurk, S. et al. HiCanu: accurate assembly of segmental duplications, satellites, and allelic variants from high-fidelity long reads. *Genome Res.* **30**, 1291–1305 (2020).
38. Sadasivan, H. et al. Accelerating Minimap2 for accurate long read alignment on GPUs. *J. Biotechnol. Biomed.* **6**, 13–23 (2023).
39. Kalikar, S., Jain, C., Vasimuddin, M. & Misra, S. Accelerating minimap2 for long-read sequencing applications on modern CPUs. *Nat. Comput. Sci.* **2**, 78–83 (2022).
40. Šošić, M. & Šikić, M. Edlib: a C/C++ library for fast, exact sequence alignment using edit distance. *Bioinformatics* **33**, 1394–1395 (2017).
41. Groot Koerkamp, R. & Ivanov, P. Exact global alignment using A* with chaining seed heuristic and match pruning. *Bioinformatics* **40**, btae032 (2024).
42. Kolmogorov, M. et al. metaFlye: scalable long-read metagenome assembly using repeat graphs. *Nat. Methods* **17**, 1103–1110 (2020).
43. Feng, X., Cheng, H., Portik, D. & Li, H. Metagenome assembly of high-fidelity long reads with hifiasm-meta. *Nat. Methods* **19**, 671–674 (2022).
44. Bertrand, D. et al. Hybrid metagenomic assembly enables high-resolution analysis of resistance determinants and mobile elements in human microbiomes. *Nat. Biotechnol.* **37**, 937–944 (2019).
45. Vaser, R. & Šikić, M. Time- and memory-efficient genome assembly with Raven. *Nat. Comput. Sci.* **1**, 332–336 (2021).
46. Stanojević, D., Li, Z., Bakić, S., Foo, R. & Šikić, M. Rockfish: A transformer-based model for accurate 5-methylcytosine prediction from nanopore sequencing. *Nat. Commun.* **15**, 5580 (2024).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026

Methods

HERRO is a deep learning-based error-correction tool for ONT long reads, developed in collaboration with ONT. ONT subsequently integrated the HERRO algorithm into their Dorado basecalling framework as dorado correct. Although dorado correct is based on the HERRO algorithm, the two implementations are now maintained independently, and future algorithmic improvements or optimizations in either tool will not necessarily be reflected in the other. All analyses in this paper were performed using the standalone HERRO implementation.

Data preprocessing

Data preprocessing consists of read basecalling, adapter trimming, read splitting on middle adapters and filtering out short and low-quality reads. All datasets were basecalled using Dorado (<https://github.com/nanoporetech/dorado>), with the model version depending on the pore type. R9.4.1 data was basecalled using the dna_r9.4.1_e8_sup@v3.6 model, R10.4.1 4 kHz data using the dna_r10.4.1_e8.2_400bps_sup@v4.1.0 model and R10.4.1 5 kHz data using the dna_r10.4.1_e8.2_400bps_sup@v4.3.0 model. Adapter trimming is done using Porechop⁴⁷, wherein we recognize and remove adapter sets with at least one match over 95%. Next, we detect middle adapters and split reads using the split_on_adapter script from Duplex Tools (<https://github.com/nanoporetech/duplex-tools>). Dorado versions v0.5.0 or higher implement adapter trimming and splitting, making Porechop and Duplex Tools unnecessary. Finally, reads shorter than 10 kb and those with a mean base quality below 10 are discarded using SeqKit^{48,49}. Only the resulting reads will be used in the error-correction process. Read filtering by length and quality was used in our experiments and is recommended but is not built into the main pipeline. Additional details and the preprocessing script are available in Supplementary Information 1 and 2.

Feature generation

To generate features, we first perform all-versus-all alignment using minimap2^{22,50}. If multiple alignments between two reads exist, we keep only the first one, assuming that alignment is optimal. We perform error correction on each read independently. The read that will be corrected is named target read, while the reads aligned with it are named query reads.

Initially, we divide a target read into chunks by defining non-overlapping windows of length $L_T = 4,096$ base pairs (bp). We extract the corresponding segments from the query reads for each window based on the alignments (CIGAR string). Any query segment that doesn't cover the entire window is discarded. Moreover, a query segment for a window is also discarded if any insertion or deletion is longer than 50 bp. Next, we sort segments by alignment accuracy and retain only the top 30 segments. All segments are retained if there are fewer than 30 for a window. For each window, we define two tensors: $B \in \mathbb{R}^{31 \times L}$ and $Q \in \mathbb{R}^{31 \times L}$, corresponding to the stacked pairwise bases and stacked pairwise base qualities, respectively. The first row in both tensors corresponds to the target read, while the subsequent rows contain the sorted segments in decreasing order. If fewer than 30 segments exist, we insert a special 'no alignment' token (.) in the remaining rows.

Moreover, we define a special gap token (*) to denote insertions with respect to the target read. This results in a total tensor length $L = L_T + \sum_{i=1}^{L_T} L_i^i$ where L_i^i is the length of the longest insertion at the target position i . Each column, which can be either a target or inserted position, is later called independently. Information about the alignment strand is given implicitly; bases from different strands are encoded differently. There are 11 tokens: 5 bases (A, C, G, T and *) for each strand (forward and reverse) plus a no alignment token. Base qualities for the insertions are set to 0.

In addition to features, we also extract information about informative position candidates. We define a position as an informative position

candidate if at least two different bases (including gap) are supported by a minimum of three reads at that specific position. This requirement introduces a lower bound on the detectable variant allele frequency, which depends on sequencing depth (for example, approximately 10% at 30× coverage, 3% at 100×). As a result, HERRO is not suitable for identifying low-frequency variants, especially at low coverage. Positions that do not meet this criterion are considered to be non-candidate positions. The deep learning model predicts only the target bases for informative position candidates.

Deep learning model and training

Model architecture. HERRO deep learning model consists of three parts (Fig. 1). In the first part, we update information for each base based on its local neighbourhood before pooling information from each base at the same position to obtain a vector representation for each position. Initially, the base tensor is embedded using an embedding layer, resulting in the embedded bases $E_B \in \mathbb{R}^{31 \times L \times 6}$. This tensor and the quality base tensor are concatenated into an input tensor $I \in \mathbb{R}^{31 \times L \times 7}$, processed through two convolutional blocks. Each convolutional block consists of 1D convolution, batch normalization and rectified linear unit (ReLU). The output of the first stage is the intermediate tensor $H^{(1)} \in \mathbb{R}^{L \times 256}$. In the second part, we focus only on representations from informative position candidates, updating them based on interactions with other candidates. By concentrating only on the candidates, we process only a fraction of all positions, drastically reducing the computational demands. The tensor $H^{(2)} \in \mathbb{R}^{S \times 256}$, containing representations of informative position candidates is fed into a Pre-LN Transformer encoder^{51,52}. The output from the encoder is passed to the final part of the model, which consists of two classification heads: the base prediction head and the informative position prediction head. The base prediction head, consisting of a fully connected layer followed by a softmax layer, produces base probabilities $Y_B \in \mathbb{R}^{S \times 5}$ for every candidate position. For each candidate position, it predicts whether the correct base is one of the canonical bases (A, C, G, T) or a gap symbol (*). The informative position prediction head, defined as a fully connected layer followed by a sigmoid, outputs the probabilities $Y_I \in \mathbb{R}^{S \times 1}$, which denotes whether each candidate is an informative position. A candidate position is deemed informative if different haplotypes have different bases at that position.

Model training. We perform multi-task learning to find the parameters of the optimal model. The total model loss for one window is the sum of the losses from each candidate position. The total loss for the candidate position i , where $i \in \{1, 2, \dots, S\}$, is a sum of two losses: base prediction loss and informative position prediction loss and is given by:

$$L = L_B + L_I = \text{CE}(Y_B^i, b^i) + \text{BCE}(Y_I^i, \mathbb{1}_{b_{h_1}^i \neq b_{h_2}^i})$$

where CE is the cross-entropy loss, BCE is the binary cross-entropy loss, Y_B^i represents the predicted base probabilities, b^i is the true base, Y_I^i denotes the predicted probability that a candidate position is informative, and $\mathbb{1}_{b_{h_1}^i \neq b_{h_2}^i}$ is the indicator function that equals 1 if the bases on haplotypes h_1 and h_2 are different, and 0 otherwise.

We trained the model on four A100 GPUs for a maximum of one million steps. The batch size was 128, so the model saw 128 million examples. Validation was performed every 2,000 steps, and we selected the model with the highest validation F1 score for the base prediction task. The Adam optimizer was used for training with default $\beta_1 = 0.9$ and $\beta_2 = 0.999$ parameters. We used a cosine learning rate scheduler with an initial learning rate of 3×10^{-4} . Weight decay was set to 0.1 and gradients were clipped using the ℓ^2 -norm with a maximum norm value of 1. The dropout value in transformer was set to 0.1. The complete list of hyperparameters can be found in Supplementary Table 11.

Training dataset. We used a portion of the HG002 dataset internally sequenced by ONT for training (HG002 internal), specifically chromosomes 1–2 and 17–18. Chromosome 20 was used for validation. The remaining chromosomes were excluded from training and model selection. The reads were first aligned to the recently published high-quality assembly using minimap2. This diploid assembly allowed us to assign the chromosome and haplotype to each read. Only reads with primary alignments to the specified chromosomes and mapping quality of 2 or higher were processed further, as previously described. To identify correct bases and informative positions, we mapped both contigs (haplotypes) corresponding to the assigned chromosome to each read. We extracted the correct base from the assigned haplotype for each position on a read. If the bases differed between the two haplotypes at a particular position, that position was labelled informative. If there was no mapping for the non-assigned haplotype, all positions were automatically labelled as non-informative.

Consensus

After obtaining predictions from the deep learning model for the candidate positions, we determine a consensus sequence for each window. For non-candidate positions, we use simple majority voting. If two or more bases (including gaps) occur with the same frequency and one of them is a target base, we copy the target base; otherwise, the base is chosen randomly. For the candidate positions, we select the base with the highest predicted probability. Consecutive windows are stitched into a single sequence. If a window contains fewer than two alignments, the window is discarded, and the read is split. The resulting sequences are written to a FASTA file.

Analysis methods

Generating assemblies with Hifiasm. We ran Hifiasm v0.19.8 to generate assemblies for all corrected datasets, subsampling them to 35× coverage if they exceeded 38×. The corrected reads were chopped to 30 kb and filtered to retain only those longer than 10 kb. According to our experiments, this was necessary because Hifiasm did not perform optimally with longer reads as high-accuracy inputs. This could be related to the nature of its algorithm^{27,28}. Reads that were trimmed (using porechop and duplex_tools on the fastq or trimmed during basecalling by Dorado v0.5.0 and above) but not yet corrected and that were 50 kb or longer were used as the UL input for Hifiasm. In the case of haploid genomes, primary assemblies are evaluated. For diploid samples, parental data were processed using yak and used with Hifiasm's built-in trio-binning mode to obtain phased assemblies. For detailed commands, versions and parameters, see Supplementary Information 3.

Generating assemblies with Verkko. Verkko was run on all corrected datasets to generate assemblies, subsampling them to 35× if their coverage was 38× or above. We generally used Verkko v2.0 by default; however, in some instances where it crashed, we used a newer version (Supplementary Information 3). For Verkko, the corrected reads were used as is, without chopping. Trimmed, uncorrected reads were used as the UL input for Verkko without filtering by length. For diploid samples, Meryl and Merqury were used to process parental data in Verkko's trio mode to generate phased assemblies. Homozygous or haploid samples were run with the –haploid option to improve the alignment of ONT reads. Refer to Supplementary Information 3 for detailed commands, versions and parameters.

Generating haploid assemblies with LJA. LJA (v0.2) was run on all homozygous or haploid corrected datasets to generate assemblies, subsampling them to 35× if they were 38× or above. Only the downsampled corrected reads were used as input for LJA. For detailed commands, see Supplementary Information 3.

Evaluation of gene completeness. We evaluated the gene completeness of assemblies using three different methods—asmgene, BUSCO^{53,54} and compleasm⁵⁵.

For asmgene, a reference assembly was used for each sample. Minimap2 2.26-r1175 was used to align cDNA sequences to the reference assembly and then to the constructed assembly (for diploid samples, each haplotype's assembly is assessed separately). We then ran asmgene from paftools 2.26-r1175 and used its outputs to calculate how many single-copy or multi-copy genes found on the reference remained single- or multi-copy on the assembly. From this, we calculated the proportion of single-copy genes on the reference that were no longer single-copy on the assembly, the proportion of single-copy genes that became duplicated, and the proportion of multi-copy genes that were no longer multi-copy by the following:

Proportion of lost single-copy genes

$$= \frac{\text{full_sgl}_{\text{reference}} - \text{full_sgl}_{\text{assembly}}}{\text{full_sgl}_{\text{reference}}}$$

Proportion of duplicated single-copy genes = $\frac{\text{full_dup}_{\text{assembly}}}{\text{full_sgl}_{\text{reference}}}$

Proportion of lost multi-copy genes

$$= \frac{\text{dup_cnt}_{\text{reference}} - \text{dup_cnt}_{\text{assembly}}}{\text{dup_cnt}_{\text{reference}}}$$

For non-human datasets, we also additionally ran BUSCO 5.6.1 and compleasm 0.2.5 to obtain a measure of gene completeness by checking for the existence of conserved single-copy orthologues in the assembly. For detailed commands and reference for each sample, see Supplementary Information 4.

Reference-based evaluation of assemblies. Minigraph 0.20-r574-dirty with paftools 2.26-r1175 was used to assess contiguity, completeness and correctness of each assembly against a reference by alignment. We first aligned assembly sequences to the reference sequences using minigraph and indexed the reference with Samtools⁵⁶ 1.19.2 (Using htlib⁵⁷ 1.19.1). Paftools then used this information to calculate metrics such as NG50 and NGA50. NG50 is calculated by summing the lengths of the assembled sequences in descending order, and NGA50 is similar but uses aligned segments of assembly sequences to the reference; both metrics identify the sequence length at the point where cumulative length equals or exceeds 50% of the genome length. The percentage of the reference sequences aligned to a sequence in the assemblies measures assembly completeness (Rcov; Supplementary Tables 1–8 and 10). Additionally, Quast 5.2.0 with minimap2 2.24-r1122 was used to evaluate assemblies against curated reference assemblies; Liftoff⁵⁸ together with stratification bed files from GIAB⁵⁹ was used to stratify evaluation by repeat status for HG002. For detailed commands and the references used, see Supplementary Information 4.

k-mers based evaluation of assemblies. We used Merqury (with Meryl) and yak to assess assemblies using short, accurate reads from the same sample. *k*-mers in the short reads were counted and compared with the *k*-mers in assembly sequences to estimate the assembly's QV (a log-scaled measure of base error rate). For diploid samples, maternal- and paternal-specific *k*-mers were derived from the short reads of the parents of the assembled sample. This approach identifies *k*-mers from both parents in the assembled sequences, allowing it to identify switch and Hamming errors. Refer to the Supplementary information 4 for detailed versions, commands and parameters.

Article

Counting T2T chromosomes in assemblies. The script from https://github.com/prasad693/Tel_Sequences was used to identify T2T sequences in the assemblies of all samples, except for *D. melanogaster*, for which telomeres do not consist of short motif repeats⁶⁰. The script searches for species-specific canonical telomeric motifs at the ends of each sequence in the assembly and attempts to align these sequences to a reference sequence. Assembly sequences that cover the full length of the reference chromosome and also have motif repeat counts exceeding a specified threshold were considered T2T. For detailed commands, see Supplementary Information 4.

Measurement of quality of reads. Reads before and after correction were aligned to a reference for samples with a high-quality reference using minimap2 2.26-r1175. The bamConcordance tool was used on the alignments to calculate Q_v , a score that is a log-scaled measure of base error probabilities. It also outputs the counts of different types of errors—mismatch, non-homopolymer insertion, non-homopolymer deletion, homopolymer insertion, and homopolymer deletion of the reads—compared to the reference. Only the records for the primary alignments were used. Yak 0.1-r69-dirty was used to estimate QV, a log-scaled measure of base error probabilities of the reads for I002C reads corrected at different coverages. Corrected reads from HG002 (at 66×), I002C (at 63×) and CHM13 (67×) were also aligned to different high-quality human assemblies to estimate the error rates of reads with respect to them. Error rates were calculated by the following:

$$\text{Error rate} = \frac{\text{mismatch} + \text{insertion} + \text{deletion}}{\text{match} + \text{mismatch} + \text{insertion} + \text{deletion}} \times 100\%$$

$$\text{Mismatch rate} = \frac{\text{mismatch}}{\text{match} + \text{mismatch} + \text{insertion} + \text{deletion}} \times 100\%$$

$$\text{Indel rate} = \frac{\text{insertion} + \text{deletion}}{\text{match} + \text{mismatch} + \text{insertion} + \text{deletion}} \times 100\%$$

Indels longer than 50 bp are typically classified as structural variants (SVs) and were therefore excluded from the analysis. Additionally, reads aligned to mitochondrial DNA or EBV were also excluded. Additionally, we used outputs from q100bench (<https://github.com/nhansen/q100bench>) to find remaining and added errors in corrected reads, by comparing the errors before and after correction, and also to evaluate homopolymer accuracies. We also used Mosdepth⁶¹ for finding regions with low coverages of reads. BEDTools⁶² was used for operating with annotation BED files which were used for stratifying analysis by different kinds of regions. For detailed commands and alignment references, see Supplementary Information 4.

Structural variant benchmarking. We also did structural variants calling benchmarking with the GIAB SV benchmark⁶³. We used Sniffles2⁶⁴ and cuteSV^{65,66} for structural variant calling and used Truvari⁶⁷ for evaluation. We used Integrative Genomics Viewer^{68–70} (IGV) for the manual inspection of cases. For detailed commands, please refer to Supplementary Information 4.

Runtime and resource assessment. The elapsed time, CPU time and peak RAM usage were measured for the all-versus-all read alignment, correction, and assembly steps. These numbers were taken from the output log for Hifiasm assembly runs and obtained with `usr/bin/time -v` for the rest. Multiple machines were used for the runs, please refer to Supplementary Information 8 for the details.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

HG02818 UL ONT data: https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=submissions/3471c92c-5b6a-4c20-9953-13e3a851debc-UCSC_HPRC_PLUS_nanopore_WashU_samples/HG02818/; HG02818 and parents' short reads data: https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=working/HPRC_PLUS/HG02818/raw_data/Illumina/; HG005 ONT data: https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=submissions/5b73fa0e-658a-4248-b2b8-cd16155bc157-UCSC_GIAB_R1041_nanopore/HG005_R1041_Sheared/; HG005 and parents' short reads data: https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=working/HPRC_PLUS/HG005/raw_data/Illumina/; HG002 UL ONT data: https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=submissions/5b73fa0e-658a-4248-b2b8-cd16155bc157-UCSC_GIAB_R1041_nanopore/HG002_R1041_UL/dorado/v0.4.0_wMods/ (only the data from the four files with 'ULCIR' or 'ULNEB' in their filenames were used); HG002 and parents' short reads data: https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=working/HPRC_PLUS/HG002/raw_data/Illumina/; *D. melanogaster* ONT data: <https://labs.epi2me.io/open-data-drosophila/>; *D. melanogaster* short reads data: Sequence Read Archive (SRA) accession SRR6702604; *A. thaliana* ONT data: SRA accession SRR29061597 (Supplementary Information 7); *A. thaliana* Illumina data: Genome Sequence Archive (GSA) accession CRA005350; *D. rerio* UL ONT data: [https://genomeark.s3.amazonaws.com/index.html?prefix=species/Danio_rerio/fDanRer17/genomic_data/ont/\(PAG68679](https://genomeark.s3.amazonaws.com/index.html?prefix=species/Danio_rerio/fDanRer17/genomic_data/ont/(PAG68679) was not used for consistency because it has a different data sampling frequency from all the rest); *D. rerio* short reads data: https://www.ncbi.nlm.nih.gov/sra/?linkname=bioproject_sra_all&from_uid=1029986; CHM13 UL ONT data: <https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=T2T/CHM13/nanopore/fast5/> (some partitions were skipped for reasons such as having too many fast5 files, see Supplementary information 6); CHM13 short reads data: SRA accession SRX1082031; HG002 v1.0.1: <https://github.com/marbl/HG002>; CHM13 v2.0: <https://github.com/marbl/CHM13>; I002C v.04: <https://github.com/lbcb-sci/I002C> or <https://github.com/LHG-GG/I002C>; CN1: <https://github.com/T2T-CN1/CN1>; *D. rerio* genome assembly used as reference: GCA_033170195.2; *A. thaliana* assembly used as reference: <https://ngdc.cnc.ac.cn/gwh/Assembly/21820/show>; *D. melanogaster* assembly used as reference: GCA_018904365.1; human cDNA used: https://ftp.ensembl.org/pub/release-111/fasta/homo_sapiens/cdna/Homo_sapiens.GRCh38.cdna.all.fa.gz; *D. melanogaster* cDNA used: https://ftp.ensembl.org/pub/release-111/fasta/drosophila_melanogaster/cdna/Drosophila_melanogaster.BDGP6.46.cdna.all.fa.gz; *A. thaliana* cDNA used: https://ftp.ensemblgenomes.ebi.ac.uk/pub/plants/release-58/fasta/arabidopsis_thaliana/cdna/Arabidopsis_thaliana.TAIR10.cdna.all.fa.gz; *D. rerio* cDNA used: https://ftp.ensembl.org/pub/release-111/fasta/danio_rerio/cdna/Danio_rerio.GRCz11.cdna.all.fa.gz; HG002 UL Q28 dataset used: https://labs.epi2me.io/gm24385_ncm23_preview/; HG002 UL ONT dataset used for training the R10.4.1 model has been submitted to NCBI under Project ID PRJNA1302743; I002C sequencing data is available at NCBI under Project ID PRJNA1150503; HG002 data used for training R9 model: https://labs.epi2me.io/gm24385_2021.05/; references used for HG005: https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/018/506/965/GCA_018506965.2_HG005_mat_hprc_f2/GCA_018506965.2_HG005_mat_hprc_f2_genomic.fna.gz and https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/018/506/945/GCA_018506945.2_HG005_pat_hprc_f2/GCA_018506945.2_HG005_pat_hprc_f2_genomic.fna.gz; references used for HG02818: https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/018/503/585/GCA_018503585.2_HG02818_mat_hprc_f2/GCA_018503585.2_HG02818_mat_hprc_f2_genomic.fna.gz and https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/018/503/575/GCA_018503575.2_HG02818_pat_hprc_f2/GCA_018503575.2_HG02818_pat_hprc_f2_genomic.fna.gz

hprc_f2_genomic.fna.gz; references used for HG01099: https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/042/027/475/GCA_042027475.1_HG01099_pat_hprc_f2/GCA_042027475.1_HG01099_pat_hprc_f2_genomic.fna.gz and https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/042/026/525/GCA_042026525.1_HG01099_mat_hprc_f2/GCA_042026525.1_HG01099_mat_hprc_f2_genomic.fna.gz; fast5 for HG002 re-basecalling (those with ULCIR or ULNEB in the filenames): https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=submissions/5b73fa0e-658a-4248-b2b8-cd16155bc157-UCSC_GIAB_R1041_nanopore/HG002_R1041_UL/; fast5 for HG01099 re-basecalling: https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=submissions/90A1F283-2752-438B-917F-53AE76C9C43E-UCSC_HPRC_nanopore.Year2/HG01099_nanopore/; Illumina reads used for Hypo-assembler (the first 627,061,028 pairs were used, total ~60×): [https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/data/indexes/AshkenazimTrio/sequence.index.AJTrio_Illumina300X_wgs_07292015_updated.HG002](https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/data/indexes/AshkenazimTrio/sequence.index.AJTrio_Illumina300X_wgs_07292015_updated.HG002; HiFi reads' assembly in Table 2: https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=submissions/53FEE631-4264-4627-8FB6-09D7364F4D3B-ASM-COMP/HG002/assemblies/verkko_1.3.1/trio/); HiFi reads' assembly in Table 2: https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=submissions/53FEE631-4264-4627-8FB6-09D7364F4D3B-ASM-COMP/HG002/assemblies/verkko_1.3.1/trio/; Duplex reads' assembly in Table 2: https://obj.umiacs.umd.edu/marbl_publications/duplex/HG002/asms/duplex_50x_30xUL_trio.tar.gz; most of the assemblies evaluated in this paper: <https://zenodo.org/records/13702708> (ref. 71) and <https://zenodo.org/records/15062959> (ref. 72); Q28 UL ONT assemblies evaluated in this paper: <https://zenodo.org/records/13703130> (ref. 73); source data are provided with this paper.

Code availability

HERRO code is available at <https://github.com/lbcb-sci/HERRO>. The pre-trained models are available at <https://zenodo.org/records/12683277> (ref. 74). Code used in this manuscript is archived at <https://zenodo.org/records/19547612> (ref. 75).

47. Wick, R. R., Judd, L. M., Gorrie, C. L. & Holt, K. E. Completing bacterial genome assemblies with multiplex MinION sequencing. *Microb. Genom.* **3**, e000132 (2017).
48. Shen, W., Sipos, B. & Zhao, L. SeqKit2: a Swiss army knife for sequence and alignment processing. *iMeta* **3**, e191 (2024).
49. Shen, W., Le, S., Li, Y. & Hu, F. SeqKit: A cross-platform and ultrafast toolkit for FASTA/Q file manipulation. *PLoS ONE* **11**, e0163962 (2016).
50. Li, H. New strategies to improve minimap2 alignment accuracy. *Bioinformatics* **37**, 4572–4574 (2021).
51. Vaswani, A. et al. Attention is all you need. in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.* Vol. 30 (eds Guyon, I. et al.) 6000–6010 (ACM, 2017).
52. Xiong, R. et al. On layer normalization in the transformer architecture. in *Proc. 37th Int. Conf. Mach. Learn.* Vol. 119 (eds Daumé, H. III & Singh, A.) 10524–10533 (PMLR, 2020).
53. Manni, M., Berkeley, M. R., Seppay, M. & Zdobnov, E. M. BUSCO: assessing genomic data quality and beyond. *Curr. Protoc.* **1**, e323 (2021).
54. Manni, M., Berkeley, M. R., Seppay, M., Simão, F. A. & Zdobnov, E. M. BUSCO update: novel and streamlined workflows along with broader and deeper phylogenetic coverage for scoring of eukaryotic, prokaryotic, and viral genomes. *Mol. Biol. Evol.* **38**, 4647–4654 (2021).
55. Huang, N. & Li, H. compleasm: a faster and more accurate reimplement of BUSCO. *Bioinformatics* **39**, btad595 (2023).
56. Danecek, P. Twelve years of SAMtools and BCFtools. *Gigascience* **10**, giab008 (2021).
57. Bonfield, J. K. et al. HTSLIB: C library for reading/writing high-throughput sequencing data. *Gigascience* **10**, giab007 (2021).
58. Shumate, A. & Salzberg, S. L. Liftoff: accurate mapping of gene annotations. *Bioinformatics* **37**, 1639–1643 (2021).
59. Dwarshuis, N. et al. The GIAB genomic stratifications resource for human reference genomes. *Nat. Commun.* **15**, 9029 (2024).

60. Mason, J. M. & Biessmann, H. The unusual telomeres of *Drosophila*. *Trends Genet.* **11**, 58–62 (1995).
61. Pedersen, B. S. & Quinlan, A. R. Mosdepth: quick coverage calculation for genomes and exomes. *Bioinformatics* **34**, 867–868 (2018).
62. Quinlan, A. R. & Hall, I. M. BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics* **26**, 841–842 (2010).
63. Zook, J. M. et al. A robust benchmark for detection of germline large deletions and insertions. *Nat. Biotechnol.* **38**, 1347–1355 (2020).
64. Smolka, M. et al. Detection of mosaic and population-level structural variants with Sniffles2. *Nat. Biotechnol.* **42**, 1571–1580 (2024).
65. Jiang, T. et al. Long-read-based human genomic structural variation detection with cuteSV. *Genome Biol.* **21**, 189 (2020).
66. Jiang, T. et al. cuteFC: re-genotyping structural variants through an accurate and efficient force-calling method. *Genome Biol.* **26**, 166 (2025).
67. English, A. C., Menon, V. K., Gibbs, R. A., Metcalf, G. A. & Sedlazeck, F. J. Truvari: refined structural variant comparison preserves allelic diversity. *Genome Biol.* **23**, 271 (2022).
68. Robinson, J. T. et al. Integrative genomics viewer. *Nat. Biotechnol.* **29**, 24–26 (2011).
69. Thorvaldsdóttir, H., Robinson, J. T. & Mesirov, J. P. Integrative Genomics Viewer (IGV): high-performance genomics data visualization and exploration. *Brief. Bioinform.* **14**, 178–192 (2013).
70. Robinson, J. T., Thorvaldsdóttir, H., Wenger, A. M., Zehir, A. & Mesirov, J. P. Variant review with the Integrative Genomics Viewer. *Cancer Res.* **77**, e31–e34 (2017).
71. Stanojević, D., Lin, D., Nurk, S., Florez de Sessions, P. & Šikić, M. Assemblies for HERRO paper (version v2) [Data set]. *Zenodo* <https://doi.org/10.5281/zenodo.13702708> (2024).
72. Stanojević, D., Lin, D., Nurk, S., Florez de Sessions, P. & Šikić, M. Assemblies for HERRO paper 2 [Data set]. *Zenodo* <https://doi.org/10.5281/zenodo.15062959> (2025).
73. Stanojević, D., Lin, D., Nurk, S., Florez de Sessions, P. & Šikić, M. Q28 ULONT Assemblies for HERRO paper [Data set]. *Zenodo* <https://doi.org/10.5281/zenodo.13703130> (2024).
74. Stanojević, D., Lin, D., Nurk, S., Florez de Sessions, P. & Šikić, M. HERRO models (version v1). *Zenodo* <https://doi.org/10.5281/zenodo.12683277> (2024).
75. Stanojević, D. et al. lbcb-sci/herro: HERRO v0.1.1. *Zenodo* <https://doi.org/10.5281/zenodo.19547612> (2026).

Acknowledgements This work was supported by the A*STAR Computational Resource Centre through the use of its high-performance computing facilities. We acknowledge M. F. Schmitz for his invaluable help with drawings. We thank the Human Pangenome Reference Consortium, the T2T Consortium, the Vertebrate Genomes Project (VGP) and the Genome in a Bottle (GIAB) Consortium for making sequencing datasets publicly available; G. Dodd for providing the *A. thaliana* dataset; S. McKenzie for his assistance with datasets; and C. Y. Tham for assistance in transferring data. We acknowledge the National Genome Research Institute (NHGRI) for funding the following grants, which are in support of creating the human pangenome reference: 1U01HG010972, 1U01HG010971, 1U01HG010961, 1U01HG010973 and 1U01HG010963 and the Human Pangenome Reference Consortium (<https://humanpangenome.org/>). D.S. discloses support for the research of this work from AI Singapore and Oxford Nanopore Technologies (grant AISG2-100E-2021-076), National Medical Research Council, Singapore (grant MOH-000649-01), National Medical Research Council, Singapore (grant OFIRG25jan-0108) and Croatian Science Foundation (grant DOK-2018-09-8714). D.L. discloses support for the research of this work from AI Singapore and Oxford Nanopore Technologies (grant AISG2-100E-2021-076). M.Š. discloses support for the research of this work from AI Singapore and Oxford Nanopore Technologies (grant AISG2-100E-2021-076). S.N. and P.F.d.S. declare support as employees of Oxford Nanopore Technologies.

Author contributions M.Š. and P.F.d.S. conceived the project. D.S. designed and implemented HERRO, with contributions from M.Š., S.N. and D.L. D.L. performed the assembly experiments and bioinformatics analysis, assisted by M.Š., D.S. and S.N. D.S., D.L. and M.Š. organized the manuscript. D.L., D.S. and M.Š. wrote the manuscript, with additional input from P.F.d.S. and S.N. M.Š. and P.F.d.S. supervised the project, and M.Š. and S.N. provided mentorship and support throughout the project.

Competing interests Oxford Nanopore Technologies and AI Singapore jointly funded the AI-driven De Novo Diploid Assembler project, which resulted in this manuscript. M.Š. has received travel funds to speak at events hosted by Oxford Nanopore Technologies. S.N. and P.F.d.S. are Oxford Nanopore Technologies employees. The other authors declare no competing interests.

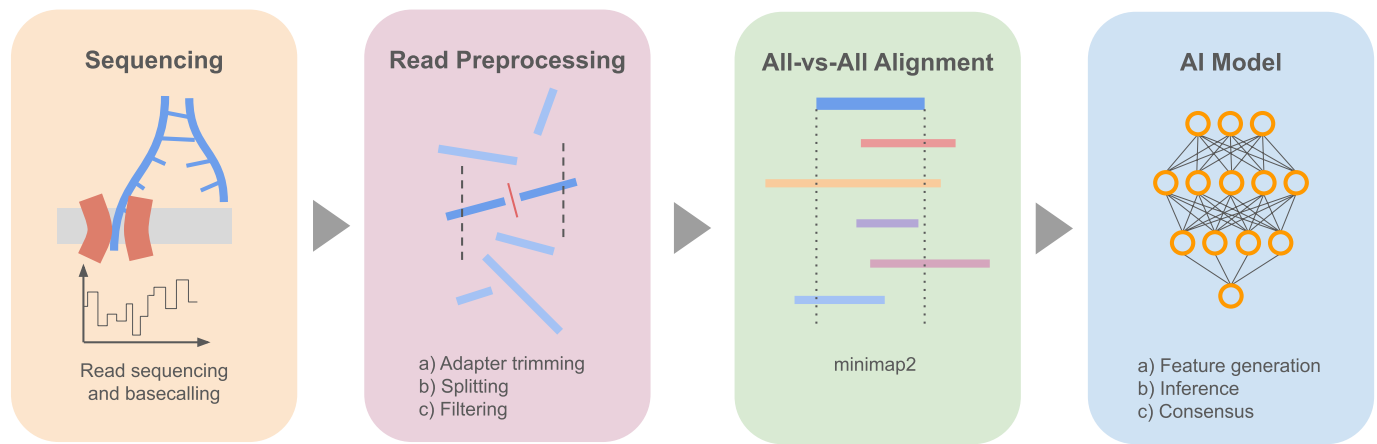
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-026-10563-y>.

Correspondence and requests for materials should be addressed to Mile Šikić.

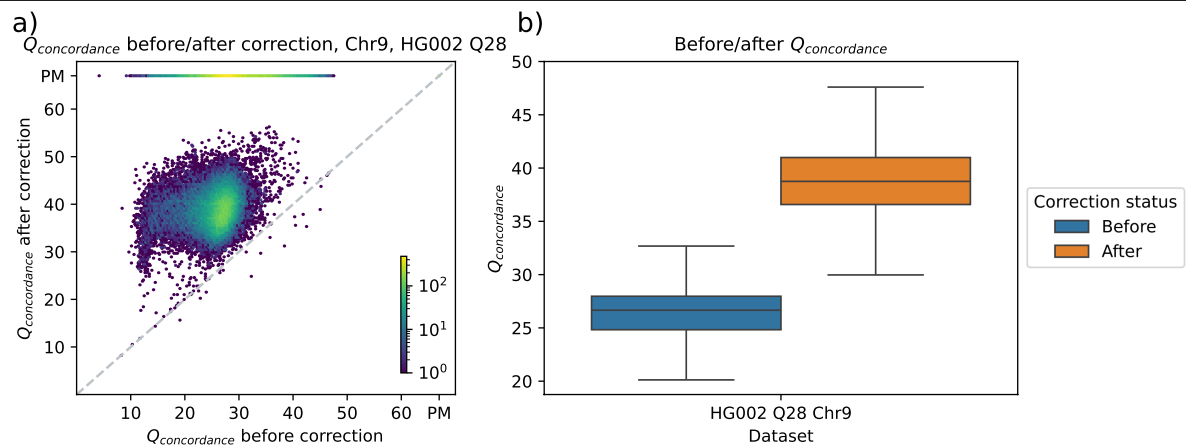
Peer review information Nature thanks the anonymous reviewers for their contribution to the peer review of this work. Peer reviewer reports are available.

Reprints and permissions information is available at <http://www.nature.com/reprints>.



Extended Data Fig. 1 | Overview of the HERRO method pipeline. The input to the pipeline are ONT sequencing reads. After trimming adapters and splitting at middle adapters, short and low-quality reads are filtered out. Next, all-vs-all alignments are performed using minimap2. In feature generation, we chunk the target read and its alignments into multiple windows. For each window, we define two matrices representing stacked bases and base qualities. These

windows are then processed by a deep-learning model. To predict the correct base for the candidate positions, we use predictions from the model. For non-candidate positions, we apply simple majority voting. The final error-corrected read is obtained by stitching together the consensus sequence from each window.



Extended Data Fig. 2 | Q_c before and after correction for Q28 ultra-long HG002 reads. a) A hexbin plot representing the data density for specific pairs of Q_c values before and after correction for reads aligned to paternal and maternal chromosome 9 of the HG002 assembly v1.0.1 ($n = 82,092$). The color intensity indicates the density of data points in each hexagonal bin on a logarithmic scale. Bins above the dashed diagonal line indicate an improvement in accuracy after correction, while bins below it denote a reduction. The “PM”

at the end of each axis stands for “Perfect Match”. b) This plot shows the distribution of Q_c before and after correction ($n = 66,359$). Alignments that perfectly match the reference were excluded. Data in the box plots are presented as follows: the centre line indicates the median, the bounds of the box represent the first and third quartiles (Q1 and Q3), and the whiskers extend to the minimum and maximum values within 1.5 times the interquartile range (IQR). Outliers are not shown due to the large number of data points.

Extended Data Table 1 | Assembly accuracy and T2T continuity for low-heterozygosity genomes

Assembly	Haplotype	Size (Gb)	NG50 (Mb)	NGA50 (Mbp)	QV	Single-copy genes lost, duplicated, multi-copy genes lost (%)	Number of T2T contigs
CHM13, ONT R9.4.1, LJA, HERRO	Pseudo-Haploid	3.052	93	92	47.52	0.04,0.009,0.55	4(4)
D. rerio, ONT, LJA, HERRO	Inbred	1.451	56	56	53.26	4.79,0.22,22.29	7
D. rerio, ONT, Hifiasm (ONT), uncorrected	Inbred	1.463	59	57	52.43	4.92,0.34,21.99	15
A. thaliana, ONT (non-UL), Hifiasm, HERRO	Inbred	0.133	26	26	48.43	0.029,0.004,0.87	3
A. thaliana, ONT (non-UL), Hifiasm (ONT), uncorrected	Inbred	0.136	26	26	48.09	0.054,0.012,1.75	3
D. melanogaster, ONT (non-UL), Hifiasm, HERRO	Inbred	0.164	24	22	45.01	0.0007,0.0006,0.05	N.A.
D. melanogaster, ONT (non-UL), Hifiasm (ONT), uncorrected	Inbred	0.158	22	22	39.31	0.06,0.04,8.97	N.A.

For each sample and correction method, we report NG50, NGA50, QV, single-copy genes lost/duplicated, multi-copy genes lost, and counts of T2T contigs. T2T continuity is assessed against the respective high-quality reference genome for each species. The first column specifies the sample, input read type, assembler, and error correction method applied; rows labelled Hifiasm (ONT) use Hifiasm's integrated internal correction, while rows labelled HERRO use explicit pre-assembly correction with HERRO. All assemblies based on HERRO-corrected reads used HERRO v0.1.0. Hifiasm (ONT) v0.25 is shown for comparison. See Methods and Supplementary Tables 5–8 (full results for HERRO-corrected read assemblies) and Supplementary Tables 5A–8A (full results for assemblies not using HERRO-corrected reads) for tool versions, coverage details, telomere criteria, and run-specific notes.

Extended Data Table 2 | Quast evaluation results of assembled T2T contigs of chromosomes X and Y

Assembly	Genome fraction (%)	Misassemblies	Mismatches per 100kbp	Total mismatches	Indels per 100kbp	Total Indels	Indel <=5 bp	Indel >5bp
ChrX, trio	100	2	0.19	286	8.43	13018	12672	346
ChrY, trio	99.98	5	1.18	736	4.08	2544	2410	134
ChrX, no-trio	100	2	0.17	270	8.43	13007	12656	351
ChrY, no-trio	99.98	5	1.16	724	4.07	2539	2405	134

Assembled T2T contigs of chromosomes X and Y using corrected HG002 reads were evaluated against HG002 v1.0.1 using Quast. The reads were corrected at 66x coverage, assembled with Verkko, with and without using parental data, respectively. All the misassemblies are relocations. The HERRO version for this analysis was v0.1.

Extended Data Table 3 | The aggregated error rates of HERRO-corrected reads with respect to different assemblies

Corrected reads dataset	Reference	Mean error rate (%)	Mean mismatch rate (%)	Mean indel rate (%)	Median error rate (%)	Percentage mapped (%)
HG002	HG002	0.023	0.0019	0.021	0.014	99.995
	CHM13	0.30	0.16	0.14	0.16	99.94
	CN1	0.36	0.19	0.17	0.17	99.94
	I002C	0.32	0.18	0.15	0.17	99.94
I002C	HG002	0.34	0.18	0.16	0.17	99.97
	CHM13	0.33	0.18	0.15	0.17	99.97
	CN1	0.34	0.18	0.16	0.17	99.97
	I002C	0.031	0.0056	0.026	0.014	99.97
CHM13	HG002	0.31	0.15	0.16	0.16	99.98
	CHM13	0.048	0.0052	0.043	0.025	99.98
	CN1	0.36	0.17	0.19	0.17	99.98
	I002C	0.33	0.16	0.16	0.17	99.98

The error rates for HERRO-corrected reads were estimated by aligning them to several high-quality assemblies. All sets of reads were aligned to CHM13 v2.0 and CN1 '0.8 curated'. HG002 reads were aligned to the HG002 v1.0.1 assembly, and I002C reads were aligned to the I002C v0.4 assembly. When aligning reads from different samples to the HG002 or I002C assemblies, only one autosomal haplotype was used (the paternal haplotype for HG002 and the maternal haplotype for I002C). Reads aligned to mitochondrial DNA or EBV sequences were excluded from the calculations. Indels longer than 50 bp were also excluded. The same measures but normalized by read length instead of alignment length can be found in Supplementary Table 18.

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

- | | |
|-------------------------------------|--|
| n/a | Confirmed |
| ☒ | ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☒ | ☐ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☒ | ☐ The statistical test(s) used AND whether they are one- or two-sided
<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
| ☒ | ☐ A description of all covariates tested |
| ☒ | ☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☒ | ☐ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☒ | ☐ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

- | | |
|-----------------|---|
| Data collection | All evaluation datasets were obtained from publicly accessible sources, which are detailed in the Data Availability section. The training data, supplied by Oxford Nanopore Technologies (ONT), is available upon request for peer-review purposes. |
| Data analysis | The complete commands and program versions used are detailed in the Supplementary Information file. |

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

HG02818 ULONT data: https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=submissions/3471c92c-5b6a-4c20-9953-13e3a851debc--UCSC_HPRC_PLUS_nanopore_WashU_samples/HG02818/ ; HG02818 and parents' short reads data: https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=working/HPRC_PLUS/HG02818/raw_data/Illumina/ ; HGO05 ONT data: https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=working/HPRC_PLUS/HG02818/raw_data/Illumina/

prefix=submissions/5b73fa0e-658a-4248-b2b8-cd16155bc157--UCSC_GIAB_R1041_nanopore/HG005_R1041_Sheared/ ; HG005 and parents' short reads data: https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=working/HPRC_PLUS/HG005/raw_data/Illumina/ ; HG002 ULONT data: https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=submissions/5b73fa0e-658a-4248-b2b8-cd16155bc157--UCSC_GIAB_R1041_nanopore/HG002_R1041_UL/dorado/v0.4.0_wMods/ (Only the data from the four files with 'ULCIR' or 'ULNEB' in their filenames were used) ; HG002 and parents' short reads data: https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=working/HPRC_PLUS/HG002/raw_data/Illumina/ ; D. melanogaster ONT data: <https://labs.epi2me.io/open-data-drosophila/> ; D. melanogaster short reads data: <https://www.ncbi.nlm.nih.gov/sra/SRR6702604> ; A. thaliana ONT data: <https://www.ncbi.nlm.nih.gov/sra/?term=SRR29061597> (See Supplementary Information, Additional notes on Arabidopsis thaliana data) ; A. thaliana Illumina data: <https://ngdc.cncb.ac.cn/gsa/browse/CRA005350> ; D. rerio ULONT data: https://genomeark.s3.amazonaws.com/index.html?prefix=species/Danio_rerio/fDanRer17/genomic_data/ont/ (PAG68679 was not used for consistency because it has a different data sampling frequency from all the rest) ; D. rerio short reads data: https://www.ncbi.nlm.nih.gov/sra/?linkname=bioproject_sra_all&from_uid=1029986 ; CHM13 ULONT data: <https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=T2T/CHM13/nanopore/fast5/> (some partitions were skipped due to reasons such as having too many fast5 files, see Supplementary information: CHM13 data partitions used) ; CHM13 short reads data: <https://www.ncbi.nlm.nih.gov/sra/?term=SRX1082031> ; HG002 v1.0.1: <https://github.com/marbl/HG002> ; CHM13 v2.0: <https://github.com/marbl/CHM13> ; I002C v.04: <https://github.com/lbcb-sci/I002C> or <https://github.com/LHG-GG/I002C> ; CN1: <https://github.com/T2T-CN1/CN1> ; D. rerio assembly used as reference: https://www.ncbi.nlm.nih.gov/datasets/genome/GCA_033170195.2/ ; A. thaliana assembly used as reference: <https://ngdc.cncb.ac.cn/gwh/Assembly/21820/show> ; D. melanogaster assembly used as reference: https://www.ncbi.nlm.nih.gov/datasets/genome/GCA_018904365.1/ ; Human cDNA used: https://ftp.ensembl.org/pub/release-111/fasta/homo_sapiens/cdna/Homo_sapiens.GRCh38.cdna.all.fa.gz ; D. melanogaster cDNA used: https://ftp.ensembl.org/pub/release-111/fasta/drosophila_melanogaster/cdna/Drosophila_melanogaster.BDGP6.46.cdna.all.fa.gz ; A. thaliana cDNA used: https://ftp.ensemblgenomes.ebi.ac.uk/pub/plants/release-58/fasta/arabidopsis_thaliana/cdna/Arabidopsis_thaliana.TAIR10.cdna.all.fa.gz ; D. rerio cDNA used: https://ftp.ensembl.org/pub/release-111/fasta/danio_rerio/cdna/Danio_rerio.GRCz11.cdna.all.fa.gz ; HG002 UL Q28 dataset used: https://labs.epi2me.io/gm24385_ncm23_preview/ ; HG002 data (HG002 internal) used for training R10 model will be available for peer-reviewers upon request; Data for I002C will be available prior to publication; HG002 data used for training R9 model: https://labs.epi2me.io/gm24385_2021.05/ ; HiFi reads' assembly in Table 2: https://s3-us-west-2.amazonaws.com/human-pangenomics/index.html?prefix=submissions/53FEE631-4264-4627-8FB6-09D7364F4D3B--ASM-COMP/HG002/assemblies/verko_1.3.1/trio/ ; Duplex reads' assembly in Table 2: https://obj.umiacs.umd.edu/marbl_publications/duplex/HG002/asms/duplex_50x_30xUL_trio.tar.gz ; Most of the assemblies evaluated in this paper: <https://zenodo.org/records/13702708> ; Q28 ULONT assemblies evaluated in this paper: <https://zenodo.org/records/13703130>

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

Use the terms *sex* (biological attribute) and *gender* (shaped by social and cultural circumstances) carefully in order to avoid confusing both terms. Indicate if findings apply to only one sex or gender; describe whether sex and gender were considered in study design; whether sex and/or gender was determined based on self-reporting or assigned and methods used. Provide in the source data disaggregated sex and gender data, where this information has been collected, and if consent has been obtained for sharing of individual-level data; provide overall numbers in this Reporting Summary. Please state if this information has not been collected. Report sex- and gender-based analyses where performed, justify reasons for lack of sex- and gender-based analysis.

Reporting on race, ethnicity, or other socially relevant groupings

Please specify the socially constructed or socially relevant categorization variable(s) used in your manuscript and explain why they were used. Please note that such variables should not be used as proxies for other socially constructed/relevant variables (for example, race or ethnicity should not be used as a proxy for socioeconomic status). Provide clear definitions of the relevant terms used, how they were provided (by the participants/respondents, the researchers, or third parties), and the method(s) used to classify people into the different categories (e.g. self-report, census or administrative data, social media data, etc.) Please provide details about how you controlled for confounding variables in your analyses.

Population characteristics

Describe the covariate-relevant population characteristics of the human research participants (e.g. age, genotypic information, past and current diagnosis and treatment categories). If you filled out the behavioural & social sciences study design questions and have nothing to add here, write "See above."

Recruitment

Describe how participants were recruited. Outline any potential self-selection bias or other biases that may be present and how these are likely to impact results.

Ethics oversight

Identify the organization(s) that approved the study protocol.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

No specific sample size was selected; the method was evaluated on the complete datasets, with the exception of the HG002 dataset.

Data exclusions	Chromosomes 1, 2, 17, and 18 were used for training, and chromosome 20 for validation. To ensure fairness, these chromosomes were excluded from the HG002 evaluation.
Replication	N.A - no experimental work was done in this study. The results are obtained using the computational method.
Randomization	N.A. - no comparisons between multiple groups that would require randomization
Blinding	No blinding was performed in this study.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.
Novel plant genotypes	Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.
Authentication	Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.